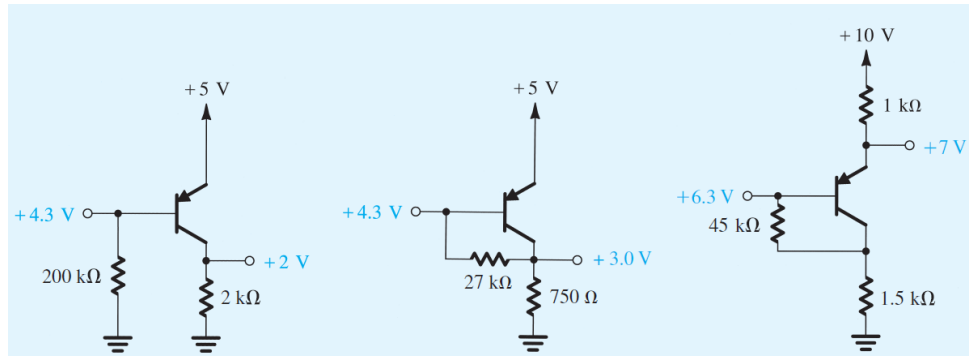
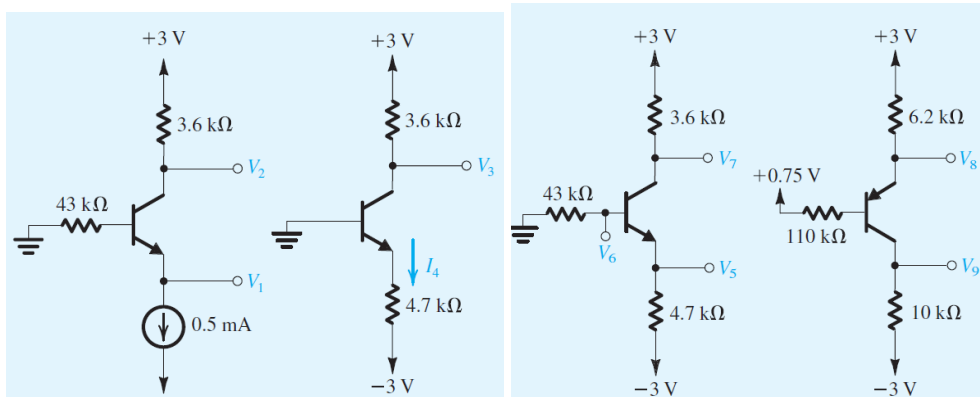


Homework 9 Topic: BJTs

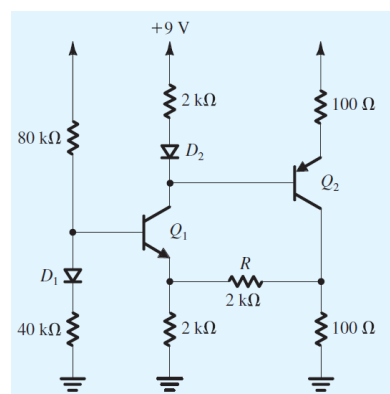
- Measurements on the circuits below produce labeled voltages as indicated. Find the value of β for each transistor.



- For the circuits in below, find values for the labeled node voltages and branch currents. Assume $\beta=100$ to be very high.



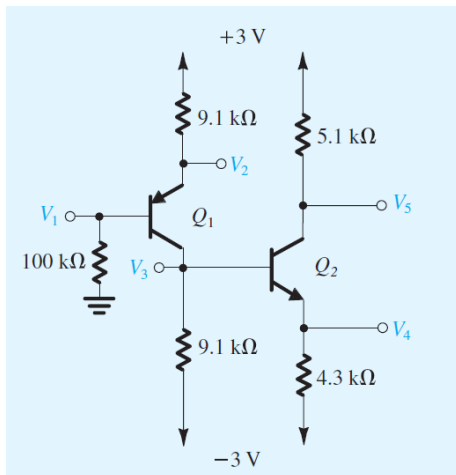
- Consider the circuit shown in figure below. Using $|V_{BE}|$ and $V_D = 0.7$ V independent of current, and $\beta = \infty$, find the voltages V_{B1} , V_{E1} , V_{C1} , V_{B2} , V_{E2} , and V_{C2} , initially with R open-circuited and then with R connected. Repeat for $\beta = 100$, with R open-circuited initially, then connected.



4. For the circuit shown below, find the labeled node voltages for:

a. $\beta = \infty$

b. $\beta = 100$



5. Given the expression

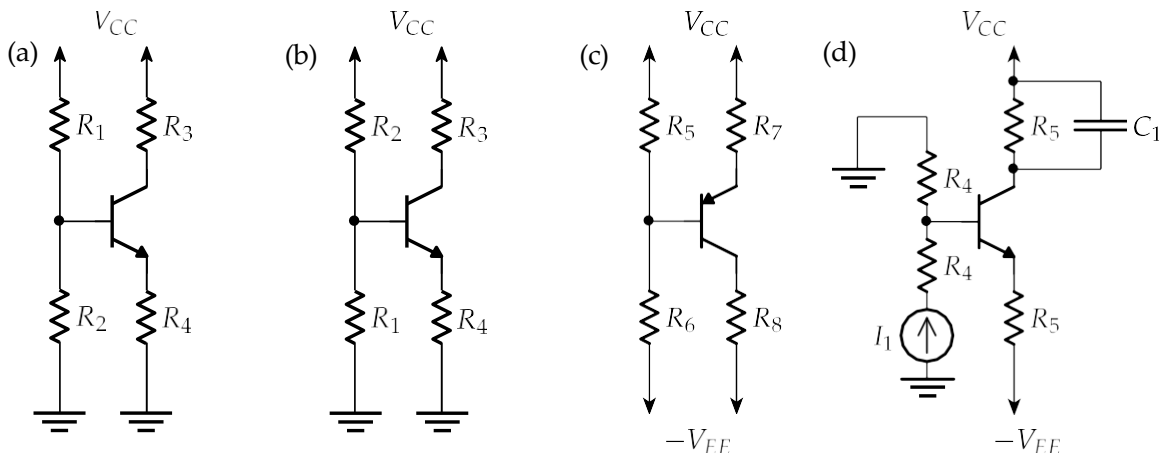
$$H(j\omega) = \frac{-\left(\frac{\omega}{100}\right)^2}{\left(1 + j\frac{\omega}{100}\right)\left(1 + j\frac{\omega}{1,000}\right)\left(1 + j\frac{\omega}{10,000}\right)}$$

- Find the starting slope of the Bode plot,
- Find the mid-band or pass-band gain,
- Determine the total phase shift,
- Plot the straight-line approximation for the magnitude and phase, and
- Plot the complete transfer function using your favorite software (phase and magnitude).
- What happened to the j in the numerator?

The goal is for you to be able to find parts (a)-(c) by examining the expression so you can use this information to predict what your sketched Bode plot is going to look like. Software should let you confirm that your solution is correct, but don't expect them to look exactly the same (the straight-line approximation is only an approximation).

6. For the following circuits, find the base, emitter, and collector voltages and currents. Use

$R_1 = 4.7 \text{ k}\Omega$, $R_2 = 1 \text{ k}\Omega$, $R_3 = 1 \text{ k}\Omega$, $R_4 = 500 \Omega$, $R_5 = 10 \text{ k}\Omega$, $R_6 = 100 \text{ k}\Omega$, $R_7 = 2.2 \text{ k}\Omega$, $R_8 = 2.7 \text{ k}\Omega$, $C_1 = 1 \mu\text{F}$, $I_1 = 10 \text{ mA}$, $V_{CC} = +12 \text{ V}$, $-V_{EE} = -12 \text{ V}$, and $\beta = 300$.



7. For the following circuits, find the base, emitter, and collector voltages and currents. Use

$R_1 = 1\text{ k}\Omega$, $R_2 = 1\text{ k}\Omega$, $R_3 = 3.3\text{ k}\Omega$, $R_4 = 5\text{ k}\Omega$, $R_5 = 10\text{ k}\Omega$, $R_6 = 1\text{ k}\Omega$, $I_1 = 1\text{ mA}$, $V_{CC} = +10\text{ V}$, $-V_{EE} = -10\text{ V}$, and $\beta = 150$. Use a constant voltage drop model for the diodes with a forward bias potential of 0.7 V .

